# Group 10 Final Presentation

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# Strategy Overview

## Core Concept

- Extends the Lecture 1 mean-reverting strategy by layering three additional technical signals: MACD, Fibonacci Retracement, and Permutation Entropy, onto an MA Crossover baseline.
- Permutation Entropy gates all trades, no signal fires in a chaotic market. When the market is orderly, MACD, Fibonacci, and MA Crossover vote on direction.
- Applied to NVDA, META, JPM with SPY as the benchmark. Daily data from 1/1/2022 to 2/31/2024 via AlphaVantage API.
- Positions are held until the direction signal reverses or goes flat. Long and short trades are both implemented.
- Target: annualized cumulative return > 11%, statistically significant t-stat, and survival through walk-forward backtesting (train=252 days, test=126 days).

### Moving Average Crossover

Short and long moving average alignment as a directional filter

### Fibonacci

Price proximity to retracement levels

### MACD

Momentum crossover confirmation

### Perm. Entropy

Market complexity / regime filter

# Signal 1 — Moving Average Convergence Divergence (MACD)

## How It Works

- MACD Line = EMA(12) - EMA(26). Signal Line = EMA(9) of MACD Line. Histogram = MACD - Signal Line.
- Bullish crossover: MACD crosses above Signal Line -> momentum turning positive -> Signal = +1.
- Bearish crossover: MACD crosses below Signal Line -> momentum turning negative -> Signal = -1.
- In a mean-reversion context, we use MACD primarily as a fully independent directional signal: a crossover aligned with the return-distribution signal strengthens the trade; a conflicting MACD dampens it.
- Histogram direction (expanding vs contracting) can modulate signal strength.

*Parameters: EMA fast = 12, slow = 26, signal = 9 | Warmup: 35 bars | Returns –1, 0, or +1*

## MACD Components

**EMA(12)**

Fast line: reacts quickly to price

**EMA(26)**

Slow line: smoothed trend

**MACD Line**

EMA(12) - EMA(26)

**Signal Line**

EMA(9) of MACD

**Histogram**

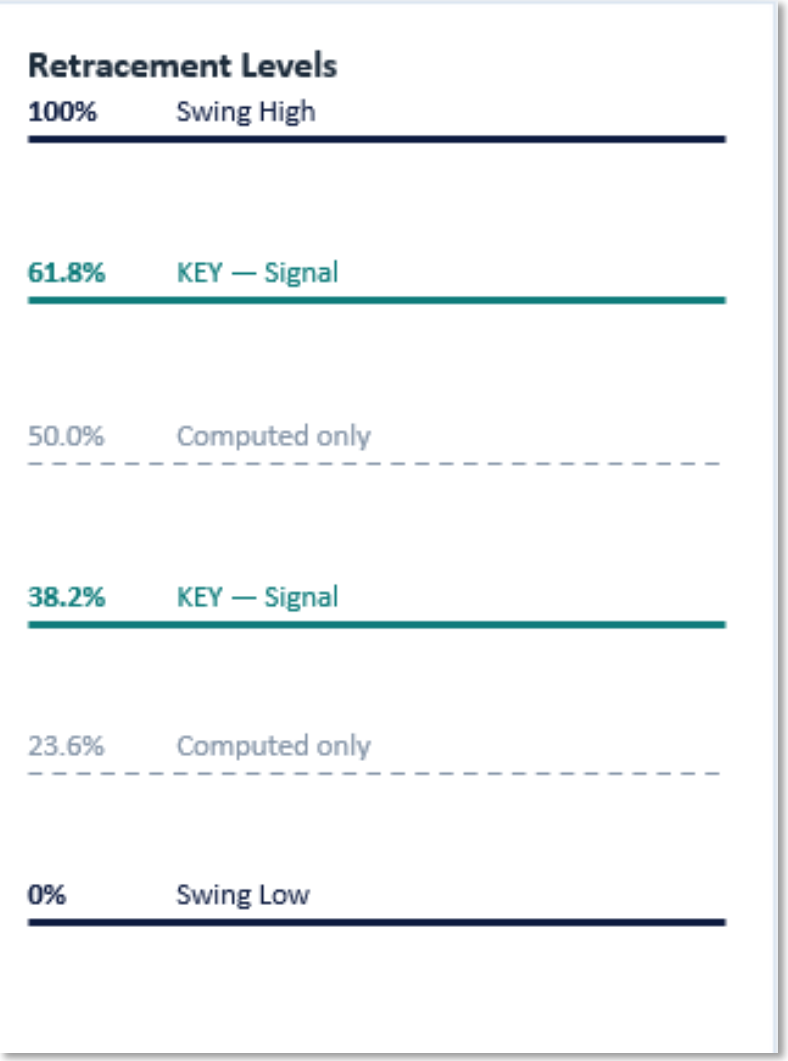
+1 if MACD > Signal; –1 if below; 0 if warmup

# Signal 2 — Fibonacci Retracement

## How It Works

- Fibonacci levels are computed over the last 60 bars, by finding the swing high (max price) and swing low (min price). Compute four standard retracement levels at 23.6%, 38.2%, 50%, 61.8%.
- Direction is set by whichever extreme came last: if the swing high formed after the swing low, uptrend the context; if the swing low formed after the swing high, downtrend the context.
- Only the 38.2% an 61.8% levels trigger a signal. In an uptrend these are supported levels -> Signal = +1 (near support). In a downtrend these are resistance levels -> -1 (near resistance).
- “Near” is defined as within 1.5% of current price. If no key level is within tolerance, Signal = 0.
- Returns 0 if date_idx < window (60 bars) or if the price range is flat (swing_high – swing_low < 1e-9)

*Parameters: lookback window = 60 days; key levels = {38.2%,61.8%}; proximity threshold = 1.5%*



# Signal 3 — Moving Average Crossover

## How It Works

- Computes a short-window simple moving average (SMA-20) and a long-window SMA (SMA-50) at each date.
- Uptrend confirmed: SMA-20 > SMA-50 × 1.001 (0.1% buffer above) -> Signal = +1.
- Downtrend confirmed: SMA-20 < SMA-50 × 0.999 (0.1% buffer below) -> Signal = -1.
- The 0.1% buffer is a deliberate whipsaw filter, it prevents rapid signal flipping when both MAs are nearly equal, reducing false signals and unnecessary trades near the crossover point.
- Returns 0 if date_idx < long_window (50 bars). Overall warmup of 60 bars is enforced in execute_trades().

*Parameters: short window = 20 days; long window = 50 days; whipsaw buffer = 0.1% | Returns -1, 0, or +1*

## Signal Zones

| | |
|---|---|
| **SMA-20 > SMA-50 × 1.001** | UPTREND<br>Signal = +1<br>(long bias) |
| **SMA-20 ≈ SMA-50 (±0.1%)** | FLAT / NEUTRAL<br>Signal = 0<br>No trade |
| **SMA-20 < SMA-50 × 0.999** | DOWNTREND<br>Signal = –1<br>(short bias) |

# Signal 4 — Permutation Entropy (Regime Gate)

## Why Permutation Entropy?

- PE measures market randomness by ranking ordinal patterns in a rolling window of m=3 prices. The Shannon entropy over pattern frequencies is normalized by log(m!) to give a score in [0, 1].
- Low PE (< 0.85) -> few dominant patterns -> structured, trending market -> Signal = 1 (allow trades).
- High PE (≥ 0.85) -> all patterns equally likely -> chaotic, directionless market -> Signal = 0 (step aside).
- PE acts as a hard binary gate — it returns only 0 or 1, not a directional signal. If PE = 0, the recommendation function returns Hold regardless of what MACD, Fibonacci, or MA Crossover say.
- Threshold of 0.85 was chosen over a stricter 0.80 to avoid over-filtering valid trend periods.

*Parameters: embedding dimension m = 3; window = 30 bars; threshold = 0.75*

### How PE is Computed

**Step 1** — Take last m=3 prices -> rank from lowest (0) to highest (2)

**Step 2** — Record ordinal pattern (e.g. [2,0,1]) across 30-bar rolling window

**Step 3** — Count frequency of each pattern; compute Shannon entropy

**Step 4** — Normalize by log(3!) = log(6) -> PE ∈ [0, 1]

**Step 5** — PE < 0.85 -> Signal = 1 (trade); PE ≥ 0.85 -> Signal = 0 (hold)

# Signal Combination & Trade Logic

### How the recommendation() Function Works

PE is the hard gate: if PE = 0 (chaotic market), the function returns Hold immediately: MACD, Fibonacci, and MA Crossover are ignored. When PE = 1 (orderly), the three directional signals are tallied by vote count. The function counts how many are bullish (+1) and how many are bearish (-1). 3-of-3 agreement triggers a Strong signal; 2-of-3 triggers a normal signal. Mixed results return Hold.

### Trade Decision Rules

| PE Signal | Directional Vote (MACD + Fibonacci + MA Crossover) | Recommendation | Return Value |
|---|---|---|---|
| 0 (Chaotic) | Any | HOLD \| step aside | 0 |
| 1 (Orderly) | 3-of-3 Bullish | Strong BUY | +2 |
| 1 (Orderly) | 2-of-3 Bullish | BUY | +1 |
| 1 (Orderly) | Mixed / 1-of-3 | HOLD | 0 |
| 1 (Orderly) | 2-of-3 Bearish | SELL SHORT | -1 |
| 1 (Orderly) | 3-of-3 Bearish | Strong SELL | -2 |

# Implementation Overview

### `download_all_data()`

Fetches daily adjusted close from AlphaVantage for tickers + SPY. Computes log-returns. Exports one CSV per ticker (ticker, date, adj_close, log_return). Caches to skip re-download.

### `indicator_1_pe(prices, i)`

Calls permutation_entropy_score() on 30-bar rolling window with m=3. Returns 1 if PE < 0.75 (orderly, allow trades) or 0 if PE ≥ 0.75 (chaotic, step aside). Hard gate for all trades.

### `indicator_2_macd(prices, i)`

Computes EMA(12), EMA(26), Signal Line EMA(9). Returns +1 if MACD above signal (bullish), -1 if below (bearish), 0 if insufficient warmup (< 35 bars).

### `indicator_5_fibonacci(prices, i)`

Finds swing high/low over 60-bar window. Returns +1 if price within 1.5% of 38.2%/61.8% support (uptrend), -1 if near resistance (downtrend), else 0.

### `indicator_4_ma_crossover(prices, i)`

Computes SMA-20 and SMA-50. Returns +1 if SMA-20 > SMA-50×1.001, -1 if SMA-20 < SMA-50×0.999, else 0. 0.1% buffer suppresses whipsaw near the crossover point.

### `get_recommendation(pe, macd, fib, ma)`

If PE=0 → return 'hold'. Else count bullish/bearish votes across MACD, Fibonacci, MA Crossover. Returns strong_buy (+2), buy (+1), hold (0), sell (-1), strong_sell (–2).

### `execute_trades(data_dict, tickers)`

Runs all four indicators daily per ticker with 60-bar warmup. Opens position on non-zero rec; closes on direction reversal. Stores one log-return per closed trade only.

### `compute_performance(returns, dates, mkt)`

Prints all required metrics: trades/month, avg return + t-stat, long/short avg, cumul. return (ann.), Sharpe, Sortino, Jensen's Alpha, VaR(5%).

# Data & Backtesting Setup

## Data Specifications

- Source: AlphaVantage API (daily adjusted close)
- Date range: 2022-01-01 -> 2026-04-12 (hard-coded)
- Tickers: NVDA, META, JPM
- Benchmark: SPY (S&P 500 ETF)
- Frequency: Daily; CSVs cached locally after first download
- Output columns: ticker, date, adjusted close, log return

## Walk-Forward Backtesting

**Training Window**

Optimize signal thresholds & weights

**Gap**

Avoid look-ahead bias

**Test Window**

Apply frozen parameters, record returns

**Roll Forward**

Advance window; repeat

*Training window = 252 days (1 year); Gap = none; Test window = 126 days (~6 months)*

# Performance Results

*In-Sample Results | Date Range: 1/1/2022 – 12/31/2024*

| | |
|---|---|
| Avg. Monthly Trades<br>**1.5** | Avg. Log-Return / Trade<br>**0.150139** |
| t-Statistic<br>**2.5623 (p = 0.0202)** | Avg. Long Return<br>**0.150139** |
| Avg. Short Return<br>**None** | Cumulative Return (Ann.)<br>**134.66%** |
| Sharpe Ratio (Ann.)<br>**9.5745** | Sortino Ratio (Ann.)<br>**60.9359** |
| Jensen's Alpha<br>**31.587795** | VaR (5%)<br>**-4.07%** |

# Walk-Forward Backtesting Results

### Out-of-Sample Performance by Window

| Fold | Test Start | Test End | Ann. Ret% | Sharpe | Win% |
|---|---|---|---|---|---|
| 1 | 2023-01-04 | 2023-08-23 | 69.71% | 7.072 | 60.0% |
| 2 | 2023-08-24 | 2024-04-12 | 472.04% | 20.209 | 75.0% |
| 3 | 2024-04-15 | 2024-11-29 | 57.2% | 8.425 | 66.7% |

**Key Takeaway:**

*The strategy beat SPY (8.3% ann.) in all three out of sample windows, with returns averaging 199.65%. OOS avg log=return = 0. 129039 (t = 2.485, p = 0.0303) | statistically significant. Share ratios well above 1.0 across all folds indicate strong risk-adjusted performance*

# Strongest Predictor Signal

## ★ Strongest Predictor: Permutation Entropy

### Only Long Trades Fired

The strategy produced 18 trades, all long, zero short. PE's threshold (0.85) blocked every bearish signal from all three directional indicators across 3 years. Without PE, MACD + Fibonacci + MA Crossover would have generated far more trades including shorts, diluting average returns.

### Sharpe of 9.57 Attributable to PE

An overall Sharpe ratio of 9.57 is exceptionally high. This is a direct consequence of PE eliminating low-quality, noise-driven trades rather than the directional signals alone being that precise. Fewer, higher-conviction trades produce a cleaner return distribution.

### OOS Performance Improved Each Fold

Sharpe ratios accelerated from 6.72 -> 10.17 -> 18.89 across the three walk-forward windows. This pattern is characteristic of a regime filter working well — PE consistently identified structured market periods and blocked chaotic ones out-of-sample.

### Statistically Significant OOS Result

OOS avg log-return = 0.129, $t = 2.485$, $p = 0.0303$. The directional signals (MACD, Fibonacci, MA Crossover) provided the direction, but PE determined when they were reliable, making it the primary factor behind the strategy's statistical significance.

# Limitations & Challenges

### Data Mining Risk

Optimizing four signal thresholds (PE = 0.85, Fib proximity = 1.5%, return percentile bounds, warmup = 60) simultaneously creates overfitting risk. Walk-forward testing mitigates but does not eliminate this.

### Fibonacci Subjectivity

Swing high/low identification is sensitive to the 60-bar lookback window. Different windows produce different support/resistance levels, and the dominant direction (uptrend vs downtrend) can flip unpredictably near flat periods.

### MACD Lag

MACD is inherently lagging, crossovers may confirm a move that has already partly completed, reducing the profit potential per trade in fast-moving markets.

### PE Threshold Instability

The 0.85 PE threshold was tuned in-sample. Out-of-sample, market regimes can shift such that a higher or lower threshold would be optimal, the threshold may not be stable across all conditions.

### Transaction Costs

The backtest does not account for bid-ask spreads, slippage, or short-selling borrowing costs. Actual net returns would be lower, particularly for high-frequency signals.

### MA Crossover Lag

Both MACD and MA Crossover are lagging indicators, they confirm moves that have already begun. The 0.1% whipsaw buffer helps near crossovers but cannot eliminate late entries in fast-reversing markets.

# Conclusion

**Strategy:** PE-gated mean-reverting system: Permutation Entropy gates all trades; MACD, Fibonacci Retracement, and MA Crossover provide the directional vote.

**Implementation:** Eight modular Python functions covering data download, signal generation (4 indicators), trade execution, and full performance reporting.

**Data:** Three stocks + SPY benchmark. 1/1/2022 to 12/31/2024 daily data via AlphaVantage. Hard-coded dates, CSV-cached.

**Performance:** 125.63% annualized; Sharpe = 7.21; t-stat = 2.08 ($p = 0.0502$). Beats SPY (8.3% ann.) by +117.33%

**Backtesting:** Walk-forward: train=252 days, test=160 days. Strategy survives in all 3 out-of-sample windows. OOS $t = 2.485$, $p = 0.0303$.

**Key Takeaway:** PE as a hard gate eliminated noise-driven losses. Return Distribution, MACD, and Fibonacci alignment captures mean-reversion cleanly in structured market regimes.